4.2.2.1 Pasteurization

Pasteurization is a heat treatment that kills part but not all the micro organisms present and involves the application of *temperatures below 100°C.* The heating, may be by means of steam, hot water, dry heat or electric currents and the products are cooled immediately after the heat treatment. *Milk is usually pasteurized.*

Pasteurized products are not sterile. They contain vegetative organisms and spores which are still capable of growth. Hence *many pasteurized foods* must be stored *under refrigeration*. Pasteurized milk can be stored for over a week under refrigeration while pasteurized milk stored at room temperature will spoil within a day.

Table 2 The Time and Temperature for the Pasteurization of Various Food Products

Food	Temperature (°C)	Duration
Milk	62.8	30 mts.
	71.7	15 sec.
Ice cream mix	71.1	30 mts.
	82.2	60–20 secs.
Grape wine	82–85	1 min.
Dried fruits	65.6–85	30–90 mts.
Bottled grape juice	76.7	30–90 mts.
Carbonated juices	65.5	30 mts.

Source: Food Science III Edition, New Age International publishers Srilalshmi. B. (2006), Chennai

4.2.2.2 Blanching

Blanching is a heat treatment like pasteurization. It is done *by dipping the products in boiling water for two to three minutes at 180°F to 190°F.* Blanching focuses on *deaerating* the product and *inactivating degradative enzymes* before further processing. Blanching is an *important step in freezing food*, as frozen foods can develop off flavour, vitamin losses and colour changes while in storage.

Blanching

- *Prevents bacterial growth.*
- *Fixes the natural colour of vegetables – holds the colour.*
- *Shrinks the product, better for filling the container.*



Blanching of tomatoes

Activity 2

1. Cut apple/banana. Expose them for 5 minutes in the air. Observe what happens.
2. Squeeze a few drops of lemon juice on a slice of apple. Does the colour change?

4.2.2.3 Canning

Canning involves the application of temperatures to food that is high enough to destroy essentially all micro organisms present. It also involves *airtight sealing* in

Chapter 4_070-087.indd 76 31-03-2018 13:21:39